How the Othello model zoo is trained and tested

1. Data (src/domains/othello.py)

Synthetic games
othello_world: random legal moves, about 19 million games

Championship games
WTHOR archive: 137,548 human games, 1977 to 2025

Three training sets
synthetic, championship, mixed (both pools together)

Rules engine replays every game
true board and legal moves after every move; illegal games dropped (none found)

split 80 / 10 / 10:
train / val / test

Tokens
one token per move: the square played (60 squares, token 1 to 60)
a game = up to 59 tokens;
passes are not tokens

the whole game fits: no window

2. Training (src/zoo/train.py, run by src/shellscripts/train_slice.sh on 4 EC2s)

One model per cell

3 architectures:
transformer, LSTM, Mamba
3 scales: 4x128, 6x256, 8x512
3 training sets: synthetic, championship, mixed
3 seeds: 0, 1, 2
= 81 models

Objective (teacher forcing)

input: moves 1 to t of a real game
output: a score for each of the 60 squares for move t+1

cross entropy at every position;
the model always reads the real history, never its own guesses

Same budget for all

5,000 steps
256 games per step
(1,280,000 games)
AdamW, lr 3e-4,
200 warmup steps,
cosine decay

Outputs

model.pt (checkpoint)

training record:
final loss, val legal move rate, sha256 of the checkpoint

3. Admission, storage and testing

Registry and admission
competence = legal move rate on 1,000 val games;
admitted if at least 0.50
(81 of 81 admitted)

Shared storage
S3: shared/group6/zoo/
one folder per model

score_slice.sh
checks sha256, then runs D1, D2, D3, D4 and fragility v1, v2
on 1,000 test games of the model's own training set

Results
one JSON per model and test
(486 files, all status ok)

Data roles never mix: train = training only;
val = admission and fitting the diagnostics;
test = every final score

Analysis (src/analysis/hypotheses.py, after tag prereg-v1.4)
master table, then H1, H2, H3 with the configuration bootstrap